AutomataStudio | New | Open | Save | PNG | Report | Traffic Light Control System | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

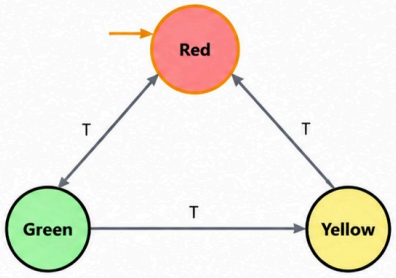
State name (e.g. q0) | Add

Red START LIVE
Green LIVE
Yellow LIVE

Set as start | Toggle accepting
Rename | Delete

Tip: drag states on the canvas. Double-click one to toggle whether it is accepting.

Traffic Light Control System | DFA




State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

| | |
|---|---|
| Deterministic (DFA) | Yes |
| Complete | Yes |
| Connected | Yes |
| ε-transitions | No |
| States / symbols | 3 / 1 |

**Unreachable states**
None

**Dead states**
None

**Missing transitions**
None

**Validation**
The automaton is complete and valid.

INPUT STRING TTTTTT | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

T T T T T T T

Step 6 / 6 — On 'T': (Yellow) → (Red)

ACCEPTED

| Step | Input Symbol | Current State | Next State |
|---|---|---|---|
| 0 | – | Red | **Red** (start) |
| 1 | T | Red | Green |
| 2 | T | Green | Yellow |
| 3 | T | Yellow | Red |
| 4 | T | Red | Green |
| 5 | T | Green | Yellow |
| 6 | T | Yellow | **Red** |

AutomataStudio | New | Open | Save | PNG | Report | Automatic Door Control System | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

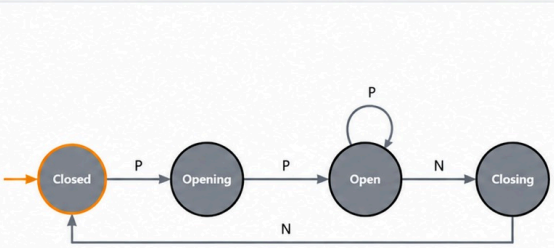
State name (e.g. q0) | Add

Closed START LIVE
Opening LIVE
Open LIVE
Closing LIVE

Set as start | Toggle accepting
Rename | Delete

Tip: drag states on the canvas. Double-click one to toggle whether it is accepting.

Automatic Door Control System | DFA




State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

| | |
|---|---|
| Deterministic (DFA) | Yes |
| Complete | Yes |
| Connected | Yes |
| ε-transitions | No |
| States / symbols | 4 / 2 |

**Unreachable states**
None

**Dead states**
None

**Missing transitions**
None

**Validation**
The automaton is complete and valid.

INPUT STRING PPNN | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭

P P N N

Step 4 / 4 — On 'N': (Closing) → (Closed)

ACCEPTED

| Step | Input Symbol | Current State | Next State |
|---|---|---|---|
| 0 | – | Closed | **Closed** (start) |
| 1 | P | Closed | Opening |
| 2 | P | Opening | Open |
| 3 | N | Open | Closing |
| 4 | N | Closing | Closed |